

Every Finite Étale Algebra of Rank at Least Three Is a Full Keller Fiber

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Abstract

Let K be a field of characteristic different from 2. Given a separable polynomial $P \in K[T]$ of degree $N \geq 3$ and a translation $a \in K$ at which the first and third Hasse derivatives of P are nonzero, we construct an explicit polynomial map $F : \mathbb{A}_K^3 \rightarrow \mathbb{A}_K^3$ with $\det DF = 1$, geometric degree N , and a distinguished full fiber naturally isomorphic to $\operatorname{Spec} K[T]/(P)$. Consequently, in characteristic zero, every finite étale K -algebra of rank at least three occurs as a full Keller fiber. For supplied presentation data the construction commutes with scalar extension; the presentation and translation chosen for an abstract finite étale algebra are noncanonical. The characteristic-zero specialization, including automatic translation, the function-field comparison, base change, and the abstract finite-étale corollary, is formalized in Lean.

1 Introduction and main result

A polynomial self-map of affine space with nonzero constant Jacobian is étale, so every one of its fibers is a finite étale scheme, possibly empty. We study the reverse problem: which finite étale algebras can occur as entire fibers of Keller maps?

The relevant condition is fullness. Let K be a field and let

$$F : \mathbb{A}_K^m \longrightarrow \mathbb{A}_K^m$$

be a Keller map. Its geometric degree is

$$\operatorname{gdeg}(F) = [K(x_1, \dots, x_m) : K(F_1, \dots, F_m)].$$

For $y \in \mathbb{A}^m(K)$, we call $F^{-1}(y)$ a *full Keller fiber* if

$$\operatorname{rank}_K F^{-1}(y) = \operatorname{gdeg}(F).$$

Thus the fiber contains every generic inverse sheet; no boundary calculation is needed once the two ranks have been identified.

Our main theorem begins with concrete presentation data.

Theorem 1.1 (Polynomial-presentation theorem). *Assume that $\operatorname{char} K \neq 2$. Let $P \in K[T]$ be separable of degree $N \geq 3$, and let $a \in K$ satisfy*

$$P^{[1]}(a)P^{[3]}(a) \neq 0,$$

where $P^{[j]}$ denotes the j -th Hasse derivative. There is an explicitly given polynomial map

$$\tilde{F}_{P,a} : \mathbb{A}_K^3 \longrightarrow \mathbb{A}_K^3$$

and an explicitly given target $y_{P,a} \in \mathbb{A}^3(K)$ such that

$$\det D\tilde{F}_{P,a} = 1, \quad \text{gdeg}(\tilde{F}_{P,a}) = N, \quad \tilde{F}_{P,a}^{-1}(y_{P,a}) \simeq \text{Spec } K[T]/(P).$$

The fiber isomorphism is natural on commutative K -algebras, every coordinate of $\tilde{F}_{P,a}$ has degree at most $6N + 2$, and the supplied (P, a) -construction commutes with extension of the ground field.

Equivalently, separability means $\text{gcd}(P, P') = 1$; this is the derivative-squarefree condition intended throughout the paper. Theorem 4.1 gives the formulas and proof. Its mechanism is a compact scalar inverse equation

$$E_{\Pi, B, C}(S) = 0$$

whose derivative is the unit used to reconstruct the source. The same factor cancels in the Jacobian. Over the generic target, the reconstruction identifies the actual source field with the field obtained by adjoining a root of E . A linear-in- C irreducibility argument then proves that this extension has degree N . At the distinguished target, $E(S) = P(a + S)$; squarefreeness removes the derivative localization and gives the literal fiber scheme. This proves fullness through

$$\text{rank}_K \tilde{F}_{P,a}^{-1}(y_{P,a}) = N = \text{gdeg}(\tilde{F}_{P,a}).$$

Every finite étale algebra over an infinite field is monogenic. In characteristic zero an admissible translation always exists. Consequently the concrete theorem gives the paper's principal abstract corollary:

Corollary 1.2. *If K has characteristic zero, every finite étale K -algebra of rank at least three occurs as a full Keller fiber of a Jacobian-one polynomial map $\mathbb{A}_K^3 \rightarrow \mathbb{A}_K^3$.*

The word “explicit” applies to the supplied polynomial presentation and admissible translation. Starting from an abstract finite étale algebra, the primitive element and the admissible translation are chosen noncanonically; the corresponding Lean construction is noncomputable and no functorial choice is claimed.

In characteristic zero, the construction realizes every rank $N \geq 3$. Together with the classical Galois case of the Jacobian theorem, due to Campbell, Razar, and Wright, it gives the possible nonzero ranks

$$1, 3, 4, \dots$$

This classification is a corollary with an external input, not part of the formalized construction.

The proof is organized in the same order as the construction. Section 2 designs the inverse equation and polynomial pullback. Section 3 proves root-to-source reconstruction, the literal fiber theorem, and the function-field comparison. Section 4 passes from a separable presentation to an abstract finite étale algebra and records scalar extension. Section 5 gives one explicit arithmetic fiber. The appendices record the coefficientwise pullback identities and the Lean correspondence.

The wider characteristic- $\neq 2$ supplied-presentation theorem is proved directly here. The current Lean development formalizes its characteristic-zero specialization; its general files still assume **CharZero**. The exact nonproperness locus, symmetric monodromy, Hilbertian specialization, and stable compositional atomicity are separate geometric and monodromy results. They are not used here.

2 The quadratic gauge from the inverse equation

Throughout this section K is a field of characteristic different from 2. A polynomial is called squarefree below in the derivative sense $\gcd(E, E') = 1$, equivalently separable over K .

The polynomial realization theorem rests on one explicit construction. Let

$$G(S) = g_1S + g_2S^2 + \cdots + g_NS^N \in K[S], \quad g_1g_3g_N \neq 0. \quad (2.1)$$

The construction is easiest to read backwards, from the inverse problem. Begin with birational coordinates (Π, S, Q) and put

$$D = 1 - SQ + \Pi S^2. \quad (2.2)$$

For the prescribed one-variable seed, define

$$G_\Pi(S) = g_1S + \Pi(g_2S^2 + g_3S^3) + \sum_{k=4}^N g_k \Pi^k S^k \quad (2.3)$$

and

$$\beta(\Pi, S) = \frac{G'_\Pi(S)/g_1 - 1 - \Pi S^2}{S}. \quad (2.4)$$

The numerator is divisible by S , so β is a polynomial. We now choose the second target coordinate to be

$$B = Q + \beta(\Pi, S). \quad (2.5)$$

This choice is the whole gauge. Indeed,

$$D = 1 - S(B - \beta(\Pi, S)) + \Pi S^2 = \frac{G'_\Pi(S)}{g_1} - BS.$$

Define the third target coordinate by

$$C = \frac{2G_\Pi(S)}{g_1} - BS^2. \quad (2.6)$$

Then the inverse equation

$$E_{\Pi,B,C}(S) = G_\Pi(S) - \frac{g_1}{2}(BS^2 + C) = 0 \quad (2.7)$$

holds by construction, and its derivative satisfies

$$D = \frac{G'_\Pi(S)}{g_1} - BS = \frac{E'_{\Pi,B,C}(S)}{g_1}. \quad (2.8)$$

This identity already gives the root-to-source reconstruction. Fix $(\pi, b, c) \in K^* \times K^2$, let s be a root of $E_{\pi,b,c}$, and suppose $E'_{\pi,b,c}(s)$ is invertible. Then the source point is recovered by the single formula

$$\boxed{\begin{aligned} d &= \frac{E'_{\pi,b,c}(s)}{g_1}, & Q &= b - \beta(\pi, s), & t &= d^{-1}, \\ x &= sd^{-1}, & y &= Q - \pi s, & q &= \pi d, \\ z &= d^2 \left(q - \frac{g_1}{g_3} y^2 (1 + 3t) \right). \end{aligned}} \quad (2.9)$$

The only localization is the inverse of the derivative unit d . The proof of Theorem 3.2 verifies over an arbitrary commutative test algebra that this formula and extraction of the marked root are mutually inverse.

The Jacobian calculation is already visible in these coordinates. At fixed Π , differentiating (2.5)–(2.6) gives

$$\det \frac{\partial(B, C)}{\partial(S, Q)} = -2D. \quad (2.10)$$

Thus any birational source chart with Jacobian D^{-1} will produce a constant Jacobian -2 . What remains is to find such a chart for which B and C pull back to polynomials.

2.1 The polynomial pullback

In source coordinates (x, y, z) , set

$$t = 1 + xy, \quad q = t^2 z + \frac{g_1}{g_3} y^2 (1 + 3t). \quad (2.11)$$

On $t \neq 0$, define

$$\Pi = tq, \quad S = \frac{x}{t}, \quad Q = y + xq. \quad (2.12)$$

Then

$$D = 1 - SQ + \Pi S^2 = t^{-1},$$

and direct differentiation gives

$$\det \frac{\partial(\Pi, S, Q)}{\partial(x, y, z)} = D^{-1}. \quad (2.13)$$

Consequently, the two Jacobian factors are

$$D^{-1} \cdot (-2D) = -2.$$

The particular polynomial q in (2.11) is chosen so that the compact coordinates (2.5)–(2.6) have no denominators after this substitution. Expanding their pullbacks gives the polynomial map $F_G = (\Pi, B, C) : \mathbb{A}_K^3 \rightarrow \mathbb{A}_K^3$:

$$\begin{aligned} \Pi &= tq, \\ B &= y + 3\frac{g_3}{g_1} xq + 2\frac{g_2}{g_1} tq + \sum_{k=4}^N k \frac{g_k}{g_1} t^2 x^{k-2} q^k, \\ C &= x(5 - 3t) - \frac{g_3}{g_1} x^3 z - \sum_{k=4}^N (k-2) \frac{g_k}{g_1} (xq)^k. \end{aligned} \quad (2.14)$$

The long formula is therefore the polynomial pullback of the simple inverse problem above, not a formula guessed in the source coordinates. Once (2.11) is fixed, the pullback is forced coefficient by coefficient. Appendix A records that expansion in full.

2.2 Motivation: the cubic seed and its quotient

For $G(S) = S + S^3$, formula (2.14) specializes coordinatewise to the map announced by Alpöge [1]. We use that map only as the cubic seed of the construction; its determinant and collision now also have an independent Isabelle/HOL formalization [5]. The geometric motivation is normalized multiplication of a marked linear factor and a quadratic factor of a binary cubic: the three marked roots give the three inverse branches, while a nonzero resultant supplies local invertibility [9]. Shaska's mixed-sign grading and quotient viewpoint gives a complementary explanation of the Jacobian cancellation on the contracted quotient [8]. Neither viewpoint replaces the derivative-unit formula (2.9), which is what identifies the complete fiber functor for an arbitrary inverse polynomial.

Thus the three-variable inverse problem is governed by the single equation $E_{\Pi,B,C}(S) = 0$, and the same derivative that controls reconstruction is exactly the factor cancelled in the Jacobian.

3 Localized fiber reconstruction and geometric degree

For $(\pi, b, c) \in K^* \times K^2$, write $\mathcal{F}_{\pi,b,c}$ for the fiber functor

$$\mathcal{F}_{\pi,b,c}(A) = \{(x, y, z) \in A^3 : F_G(x, y, z) = (\pi, b, c)\}$$

on commutative K -algebras.

Lemma 3.1 (Derivative units over test algebras). *Let $E \in K[S]$ be separable, let A be a commutative K -algebra, and let $s \in A$ satisfy $E(s) = 0$. Then $E'(s)$ is a unit of A .*

Proof. Choose $U, V \in K[S]$ with $UE + VE' = 1$. Evaluating at s gives $V(s)E'(s) = 1$. Since A is commutative, this is a two-sided inverse. $\square$

Theorem 3.2 (General localized quadratic-gauge fiber theorem). *The map F_G satisfies*

$$\det DF_G = -2, \quad \text{gdeg}(F_G) = N.$$

Its generic inverse equation is $E_{\Pi,B,C}(S) = 0$. Fix $(\pi, b, c) \in K^ \times K^2$, put $E = E_{\pi,b,c}$, and let $\overline{E'}$ be the class of E' in $K[S]/(E)$. Then*

$$\boxed{F_G^{-1}(\pi, b, c) \simeq \text{Spec}((K[S]/(E))[1/\overline{E'}])}. \quad (3.1)$$

Equivalently, for every commutative K -algebra A there is a natural bijection

$$\text{Hom}_{K\text{-alg}}((K[S]/(E))[1/\overline{E'}], A) \simeq \mathcal{F}_{\pi,b,c}(A). \quad (3.2)$$

Proof. The construction above gives the identities

$$B = Q + \beta(\Pi, S), \quad C = \frac{2G_{\Pi}(S)}{g_1} - BS^2. \quad (3.3)$$

Equations (2.13) and (2.10) give $\det DF_G = -2$ on the dense open set $t \neq 0$, and hence everywhere. For the cubic seed, this agrees with the normalized marked-factor interpretation above. The displayed calculation proves that the same cancellation persists for every seed satisfying $g_1 g_3 g_N \neq 0$.

We now identify the entire fiber. Let A be an arbitrary commutative K -algebra. An A -point (x, y, z) above (π, b, c) satisfies

$$tq = \pi.$$

Because π is a unit, both t and q are units. The chart coordinates

$$S = xt^{-1}, \quad Q = y + xq, \quad D = t^{-1}$$

are therefore defined on every point of the fiber. Equations (3.3) and (2.8) give

$$Q + \beta(\pi, S) = b, \quad E(S) = 0, \quad D = \frac{E'(S)}{g_1}.$$

Thus every source point gives a root S of E for which $E'(S)$ is a unit.

Conversely, suppose $s \in A$ satisfies $E(s) = 0$ and $E'(s) \in A^*$. Put the variables d, Q, t, x, y, q, z equal to the expressions in the boxed reconstruction formula (2.9). The derivative identity says exactly that

$$d = 1 - sQ + \pi s^2.$$

A calculation in the ring A now gives

$$1 + xy = t, \quad t^2 z + \frac{g_1}{g_3} y^2 (1 + 3t) = q, \quad tq = \pi.$$

The first identity in (3.3) gives $B = b$, and the root equation gives $C = c$. We have reconstructed a point of the original fiber.

The two procedures undo one another. Starting with a source point, one recovers

$$D = t^{-1}, \quad x = S/D, \quad y = Q - \pi S, \quad q = \pi D,$$

and then z from the definition of q . Starting with a root, the marked coordinate is again s . All formulas use ring operations and the inverse of the explicitly specified unit $E'(s)$, so they commute with homomorphisms of test algebras.

The universal property of $K[S]/(E)$ identifies its maps to A with roots of E in A . Localizing at $\overline{E'}$ imposes exactly the additional condition that $E'(s)$ be a unit. This proves (3.2), and Yoneda's lemma gives (3.1). Notice that this general localized statement requires no separability hypothesis: it removes precisely the ramified part of the inverse equation.

It remains to check the generic degree. Put

$$L = K(\Pi, B), \quad \lambda = \frac{g_1}{2},$$

and write

$$E_{\Pi, B, C}(S) = H_{\Pi, B}(S) - \lambda C \in L[C][S].$$

This polynomial has degree N in S . Its leading coefficient is $g_3 \Pi$ when $N = 3$ and $g_N \Pi^N$ when $N \geq 4$, so it is a unit of $L[C]$. Hence $E_{\Pi, B, C}$ is primitive over the UFD $L[C]$.

Assume it factors nontrivially over $L(C)[S]$. Gauss's lemma then gives a factorization in $L[C][S]$. Since the polynomial is linear in C , one factor, say $A(S)$, is independent of C . Comparing the coefficient of C shows that $A(S)$ divides the nonzero scalar $-\lambda$ in $L[S]$. Thus A is a unit, a contradiction. Therefore $E_{\Pi, B, C}$ is irreducible of degree N over $K(\Pi, B, C)$.

We finish by comparing that root extension with the actual source field. On the dense chart $t \neq 0$, the change of variables is birational:

$$K(x, y, z) = K(\Pi, S, Q).$$

The forward inclusion follows from the definitions. For the reverse inclusion put $D = 1 - SQ + \Pi S^2$. Then

$$t = D^{-1}, \quad x = \frac{S}{D}, \quad y = Q - \Pi S, \quad q = \Pi D, \quad (3.4)$$

and the defining equation for q recovers z rationally. Since

$$B = Q + \beta(\Pi, S),$$

we have $K(\Pi, S, Q) = K(\Pi, B, S)$. The second marked-line identity gives

$$C = \frac{2G_\Pi(S)}{g_1} - BS^2.$$

Because $K(\Pi, B, S) = K(x, y, z)$ has transcendence degree three and is generated by the three elements Π, B, S , those elements are algebraically independent. In particular, over $L_0 = K(\Pi, B)$, the element S is transcendental and the displayed expression for C is a nonconstant degree- N polynomial in S . Hence C is transcendental over L_0 : otherwise S , which is algebraic over $L_0(C)$, would be algebraic over L_0 . Thus Π, B, C are algebraically independent and $T = K(\Pi, B, C)$ is the actual target function field.

The evaluation homomorphism from the abstract target rational function field to the source field is therefore injective. Under it, $E_{\Pi, B, C}$ has the root S , and the reconstruction above gives

$$K(x, y, z) = K(\Pi, B, S) = T(S).$$

The irreducibility just proved makes $E_{\Pi, B, C}$ the minimal polynomial of S over T . Therefore

$$[K(x, y, z) : K(\Pi, B, C)] = [T(S) : T] = N,$$

which is exactly $\text{gdeg}(F_G) = N$. □

Corollary 3.3 (Finite-étale (squarefree) quadratic-gauge fibers). *In the setting of Theorem 3.2, suppose that $E = E_{\pi, b, c}$ is squarefree of degree N . Then the localization disappears and, for every commutative K -algebra A , there is a natural bijection*

$$\text{Hom}_{K\text{-alg}}(K[S]/(E), A) \simeq \mathcal{F}_{\pi, b, c}(A). \quad (3.5)$$

Consequently,

$$F_G^{-1}(\pi, b, c) \simeq \text{Spec } K[S]/(E),$$

and this is a full Keller fiber of rank N (in particular, finite étale).

Proof. Since E is squarefree, choose $U, V \in K[S]$ with $UE + VE' = 1$. Modulo E , the class of V is an inverse for $\overline{E'}$, so the localization in Theorem 3.2 is just $K[S]/(E)$. The natural equivalence and scheme isomorphism follow from that theorem. Its geometric-degree statement makes the rank- N fiber full. □

Corollary 3.4 (Jacobian-one effective form). *Let*

$$L(\Pi, B, C) = \left(\Pi, -\frac{B}{2}, C \right), \quad \tilde{F}_G = L \circ F_G.$$

Then

$$\det D\tilde{F}_G = 1, \quad \text{gdeg}(\tilde{F}_G) = N.$$

For every $(\pi, b, c) \in K^* \times K^2$,

$$\tilde{F}_G^{-1} \left(\pi, -\frac{b}{2}, c \right) \simeq \operatorname{Spec} \left((K[S]/(E_{\pi,b,c})) [1/\overline{E'_{\pi,b,c}}] \right).$$

If $E_{\pi,b,c}$ is squarefree of degree N , this reduces to

$$\tilde{F}_G^{-1} \left(\pi, -\frac{b}{2}, c \right) \simeq \operatorname{Spec} K[S]/(E_{\pi,b,c}).$$

In particular, a target with $b = 0$ is unchanged by the normalization. Moreover,

$$\deg \Pi \leq 7, \quad \deg B \leq 6N + 2, \quad \deg C \leq 6N,$$

so

$$\boxed{\max_i \deg(\tilde{F}_G)_i \leq 6N + 2.}$$

Proof. The linear map L has determinant $-1/2$, so the determinant, geometric degree, and localized-fiber statements follow from Theorem 3.2; the squarefree simplification follows from Corollary 3.3.

For the degree estimate, $\deg t = 2$ and $\deg q \leq 5$, hence $\deg \Pi \leq 7$. A summand of B indexed by $k \geq 4$ has degree at most

$$2 \deg t + (k - 2) + k \deg q \leq 4 + (k - 2) + 5k = 6k + 2.$$

The preceding terms have degrees at most 1, 6, and 7, so $\deg B \leq 6N + 2$. Similarly, $(xq)^k$ has degree at most $6k$, while the remaining terms of C have degree at most 4. Thus $\deg C \leq 6N$. A linear output normalization does not change these bounds. $\square$

Remark 3.5 (Exact hypotheses of the supplied-seed layer). *The root-to-source reconstruction at a target (π, b, c) uses only*

$$2 \neq 0, \quad g_1 g_3 \neq 0, \quad \pi \neq 0,$$

together with invertibility of $E'_{\pi,b,c}(s)$ at the marked root. The condition $g_N \neq 0$ records that the seed has degree N and is used in the generic irreducibility and geometric-degree calculation. Squarefreeness of the specialized inverse polynomial is used only to make the derivative invertible at every root, so that the localization in (3.1) disappears. No characteristic-zero hypothesis enters this supplied-seed layer.

4 Prescribing finite étale algebras

The construction used in the proof can be read from left to right:

$$\boxed{\text{separable } P(T) \mapsto G(S) = P(a + S) - P(a) \mapsto \tilde{F}_G^{-1}(y_{P,a}) \simeq \operatorname{Spec} K[T]/(P).}$$

The last arrow is an isomorphism of schemes, not merely a matching of K -points or a count after passing to an algebraic closure.

For $j \geq 0$, write $P^{[j]}$ for the j -th Hasse derivative, characterized by

$$P(T + S) = \sum_{j \geq 0} P^{[j]}(T) S^j. \tag{4.1}$$

Thus $P^{[1]} = P'$. In characteristic zero, $P^{[3]} = P'''/3!$, but the Hasse coefficient is the integral notion used in the supplied-presentation theorem.

Theorem 4.1 (Polynomial realization). *Let K be a field of characteristic different from 2. Let $P \in K[T]$ be separable of degree $N \geq 3$, and let $a \in K$ satisfy*

$$P^{[1]}(a)P^{[3]}(a) \neq 0.$$

Put $G(S) = P(a + S) - P(a)$, let $F_G = (\Pi, B, C)$ be the quadratic-gauge map of (2.14), and set

$$\tilde{F}_G = L \circ F_G, \quad L(\Pi, B, C) = \left(\Pi, -\frac{B}{2}, C \right).$$

Then

$$\det D\tilde{F}_G = 1, \quad \text{gdeg}(\tilde{F}_G) = N.$$

At

$$y_{P,a} = \left(1, 0, -\frac{2P(a)}{P^{[1]}(a)} \right) \tag{4.2}$$

the full Keller fiber is

$$\boxed{\tilde{F}_G^{-1}(y_{P,a}) \simeq \text{Spec } K[T]/(P)}.$$

Proof. By (4.1), the linear and cubic coefficients of G are $P^{[1]}(a)$ and $P^{[3]}(a)$, so the quadratic gauge of Section 2 is defined. At the target (4.2), its inverse polynomial is

$$G(S) - \frac{P^{[1]}(a)}{2} \left(-\frac{2P(a)}{P^{[1]}(a)} \right) = P(a + S).$$

Translation preserves separability, so Corollary 3.3 identifies the fiber with $\text{Spec } K[S]/(P(a + S))$.

Translation back to the original polynomial gives

$$K[S]/(P(a + S)) \longrightarrow K[T]/(P(T)), \quad \bar{S} \longmapsto \bar{T} - a,$$

with inverse $\bar{T} \mapsto \bar{S} + a$. The output map L fixes the chosen target, multiplies the Jacobian by $-1/2$, and changes neither geometric degree nor fiber length. Corollary 3.4 therefore gives $\det D\tilde{F}_G = 1$ and $\text{gdeg}(\tilde{F}_G) = N$. Since the displayed fiber has length N , it is full. $\square$

Corollary 4.2 (Effective and functorial form). *Under the hypotheses of Theorem 4.1,*

$$\max_i \deg(\tilde{F}_G)_i \leq 6N + 2.$$

For every commutative K -algebra R , the scheme isomorphism induces a bijection

$$\text{Hom}_{K\text{-alg}}(K[T]/(P), R) \simeq \{u \in R^3 : \tilde{F}_G(u) = y_{P,a}\},$$

natural in R .

Proof. The degree bound is Corollary 3.4. The natural bijection is the represented-fiber equivalence in Corollary 3.3, composed with the translation $K[S]/(P(a + S)) \simeq K[T]/(P)$. $\square$

Lemma 4.3 (Monogenicity). *Every finite étale algebra over an infinite field is generated by one element. Equivalently, if A has rank N , then*

$$A \simeq K[T]/(P)$$

for a separable polynomial $P \in K[T]$ of degree N .

Proof. For $\theta \in A$, let $\chi_\theta(T)$ be the characteristic polynomial of multiplication by θ on the K -vector space A , and set

$$\Delta(\theta) = \text{disc}(\chi_\theta).$$

This is a polynomial function on the affine space underlying A . After extending scalars to a separable closure, identify $A \otimes_K K^{\text{sep}}$ with $(K^{\text{sep}})^N$. In these coordinates,

$$\Delta(\theta_1, \dots, \theta_N) = \prod_{i < j} (\theta_i - \theta_j)^2,$$

so Δ is not the zero polynomial. Since K is infinite, some $\theta \in A$ satisfies $\Delta(\theta) \neq 0$. Its characteristic polynomial has N distinct geometric roots, and its minimal polynomial therefore has degree N . Hence $K[\theta] = A$, with squarefree minimal polynomial P . $\square$

Corollary 4.4 (Finite étale realization). *Let K have characteristic zero. Every finite étale K -algebra A of rank $N \geq 3$ occurs as a full Keller fiber of a Jacobian-one map*

$$\mathbb{A}_K^3 \longrightarrow \mathbb{A}_K^3$$

of geometric degree N .

More precisely, once an isomorphism $A \simeq K[T]/(P)$ and an element $a \in K$ with $P^{[1]}(a)P^{[3]}(a) \neq 0$ are supplied, the map, target, and fiber isomorphism are given by the explicit formulas of Theorem 4.1. For an abstract A , the monogenicity argument proves that such a presentation exists; it does not choose one canonically.

Proof. Lemma 4.3 supplies a squarefree presentation $A \simeq K[T]/(P)$ of degree N . In characteristic zero, $P^{[1]} = P'$ and $P^{[3]} = P'''/3!$ are nonzero polynomials. Since K is infinite, some a avoids the finite zero set of $P^{[1]}P^{[3]}$. Apply Theorem 4.1. $\square$

Corollary 4.5 (Rank classification, using the classical Galois case). *Let K have characteristic zero and let X be a finite K -scheme of positive rank N . Then X is isomorphic to a full Keller fiber of a geometric-degree- N Keller map*

$$F : \mathbb{A}_K^3 \longrightarrow \mathbb{A}_K^3$$

if and only if X is finite étale and $N \neq 2$. Consequently, the possible nonzero ranks of full Keller fibers in $\mathbb{A}_K^3$ are exactly

$$\boxed{1, 3, 4, 5, \dots}.$$

Proof. Every Keller fiber is finite étale. If it is a full Keller fiber of rank N , its map has geometric degree N . A degree-two function-field extension in characteristic zero is Galois, so the classical Galois case of the Jacobian theorem excludes $N = 2$ [3, 7, 10]. For completeness, the usual complex statement applies over an arbitrary characteristic-zero field by spreading out the finitely many coefficients, observing that generic degree is preserved by scalar extension, embedding the coefficient field in $\mathbb{C}$, and descending the polynomial inverse by faithful flatness. Conversely, the identity realizes every rank-one finite étale K -scheme, namely $\text{Spec } K$, and Corollary 4.4 realizes every finite étale scheme of rank $N \geq 3$ as a full Keller fiber. Empty and shorter special fibers are Keller fibers, but they are not full Keller fibers and hence do not alter the rank list. $\square$

4.1 Scalar extension and arithmetic transfer

Proposition 4.6 (Compatibility with extension of scalars). *Let $K \hookrightarrow K'$ be a field extension with $\text{char } K \neq 2$. For a separable $P \in K[T]$ and a translation $a \in K$ satisfying $P^{[1]}(a)P^{[3]}(a) \neq 0$, the map $\tilde{F}_G$ of Theorem 4.1, after coefficientwise extension to K' , is exactly the map obtained from the same P and a over K' . Its distinguished fiber satisfies*

$$\tilde{F}_{G,K'}^{-1}(y_{P,a}) \simeq \text{Spec}((K[T]/(P)) \otimes_K K') \simeq \text{Spec } K'[T]/(P).$$

Proof. Every formula in (2.11)–(2.14), the output normalization L , and the target (4.2) is obtained from the coefficients of P and a by field operations. The construction therefore commutes with scalar extension. Separability and the two nonvanishing Hasse coefficients are preserved by field extension. The fiber identity is the usual base-change isomorphism for a polynomial quotient, combined with Theorem 4.1. $\square$

Remark 4.7 (Supplied data versus automatic choices). *Proposition 4.6 fixes the presentation P and translation a . The passage from an abstract finite étale algebra to a primitive element, its separable minimal polynomial, and an admissible translation is noncanonical. Lean implements these existence choices using classical choice. Thus the supplied (P, a) -construction commutes with scalar extension, but no functorial automatically chosen realization is asserted.*

Corollary 4.8 (Field extensions as connected full Keller fibers). *Let K have characteristic zero. Every finite field extension L/K of degree at least three occurs as a connected full Keller fiber of a Jacobian-one polynomial map $\mathbb{A}_K^3 \rightarrow \mathbb{A}_K^3$ of coordinate degree at most $6[L : K] + 2$. Given a primitive element, its minimal polynomial, and an admissible translation, the map is explicit.*

Proof. The minimal polynomial of a chosen primitive element is separable. Characteristic zero supplies an admissible translation as in the proof of Corollary 4.4; apply Theorem 4.1. $\square$

Corollary 4.9 (Arithmetic data are preserved). *Let A be a finite étale algebra of rank at least three over a number field. Any realization supplied by Corollary 4.4 has fiber $\text{Spec } A$ itself. It therefore preserves:*

- *connectedness and field decomposition;*
- *real signature;*
- *splitting field and Galois action on geometric points;*
- *the local algebras $A \otimes_K K_v$ at every place;*
- *for any fixed integral model, factorization types away from its bad primes;*
- *the existence or nonexistence of rational and local points.*

Thus an arithmetic example built as a finite étale algebra of rank at least three can be transferred without alteration to a full Keller fiber. Proposition 4.6 gives the corresponding compatibility with local fields and other scalar extensions.

5 An explicit arithmetic fiber

Consider the squarefree quintic

$$P_5(T) = (T^3 - 19)(T^2 + T + 1) = T^5 + T^4 + T^3 - 19T^2 - 19T - 19.$$

Berend and Bilu introduced this polynomial in their study of polynomials having a root modulo every integer [2]. It has no rational root: the rational-root theorem excludes a root of $T^3 - 19$, while $T^2 + T + 1$ has discriminant -3 . On the other hand, $\text{Spec } \mathbb{Q}[T]/(P_5)$ has a point over every completion of $\mathbb{Q}$. The cubic factor has a real root. For a prime $p \notin \{2, 3, 19\}$, if $p \equiv 2 \pmod{3}$, cubing is an automorphism of $\mathbb{F}_p^\times$, so $T^3 - 19$ has a simple root modulo p ; if $p \equiv 1 \pmod{3}$, then $\mathbb{F}_p^\times$ contains a nontrivial cube root of unity, giving a simple root of $T^2 + T + 1$. At the exceptional primes, Hensel witnesses are

p	factor and approximate root	check
2	$T^3 - 19, T = 1$	$-18 \equiv 0 \pmod{2}, \quad 3 \in \mathbb{Z}_2^*$
3	$T^3 - 19, T = -2$	$v_3(-27) = 3 > 2v_3(12) = 2$
19	$T^2 + T + 1, T = 7$	$57 \equiv 0 \pmod{19}, \quad 15 \in \mathbb{Z}_{19}^*$

We now realize this algebra by the formulas above. The translation $a = 0$ is admissible, and

$$G(S) = P_5(S) - P_5(0) = S^5 + S^4 + S^3 - 19S^2 - 19S.$$

Thus $(g_1, g_2, g_3, g_4, g_5) = (-19, -19, 1, 1, 1)$. Put

$$t = 1 + xy, \quad q = t^2 z - 19y^2(1 + 3t),$$

and define $F = (F_1, F_2, F_3)$ by

$$\begin{aligned} F_1 &= tq, \\ F_2 &= 19y - 3xq + 38tq - 4t^2 x^2 q^4 - 5t^2 x^3 q^5, \\ F_3 &= 19x(5 - 3t) + x^3 z + 2(xq)^4 + 3(xq)^5. \end{aligned} \tag{5.1}$$

This is an output rescaling of the quadratic-gauge map, so

$$\det DF = -722, \quad \text{gdeg}(F) = 5.$$

At the target $y = (1, 0, -38)$, the inverse equation is

$$G(S) - \frac{-19}{2}(-2) = G(S) - 19 = P_5(S).$$

The squarefree fiber theorem therefore gives

$$F^{-1}(1, 0, -38) \simeq \text{Spec } \mathbb{Q}[T]/((T^3 - 19)(T^2 + T + 1)).$$

This full Keller fiber has points over $\mathbb{R}$ and every $\mathbb{Q}_p$, but no rational point. The displayed map, literal fiber, rank, finite étaleness, and these local and global point statements are also included in the Lean development. Composing F with the diagonal target scaling $(u, v, w) \mapsto (u, -v/38, w/19)$ gives determinant one and sends the target back to $(1, 0, -2)$.

6 Discussion

The construction preserves the finite étale algebra itself, not merely its rank or its geometric points after base change. Its load-bearing identities are the compact inverse equation, the derivative-unit reconstruction, and the function-field comparison

$$K(x, y, z) = K(\Pi, B, C)(S).$$

Together they prove, over every field of characteristic different from 2, that the distinguished literal fiber has rank equal to the actual geometric degree whenever the separable presentation and admissible translation are supplied.

This distinction also keeps the effectivity statement precise. From a supplied separable presentation $K[T]/(P)$ and admissible translation a , all map coordinates, the target, and the fiber isomorphism are explicit and commute with scalar extension. For an abstract finite étale algebra, monogenicity proves existence of such data but does not choose it canonically.

The characteristic-zero specialization of the polynomial-presentation theorem, including the actual geometric degree, and its passage to abstract finite étale algebras are formalized in Lean. Appendix B records the public endpoint declarations and the stronger hypotheses currently present in the formal files. The rank-two exclusion is deliberately separate: it uses the classical Campbell–Razar–Wright Galois case rather than the formalized construction.

Relation to earlier work. To my knowledge, no earlier source states that every finite étale K -algebra of rank at least three occurs as a full Keller fiber in $\mathbb{A}_K^3$. This priority statement is necessarily qualified: searches can miss unpublished work or a theorem stated in different language. Gallagher gives explicit all-degree Keller families with split rational fibers [4]; Migus uses available all-degree constructions in the real non-dense-image problem [6]; and Shaska studies the grading and quotient geometry of the cubic seed [8]. These results have a different quantifier order from the prescribed-algebra theorem here.

Provenance and AI assistance. The cubic seed was announced by Alpöge, who credited Akhil Mathew for suggesting the problem and Fable for producing the example [1]. Its determinant and collision have an independent Isabelle/HOL formalization [5]. The present paper uses that map only as the cubic seed of the all-degree construction.

I used generative AI tools during exploration, symbolic-checker development, proof review, literature search, Lean formalization, and manuscript editing. I checked the stated mathematical claims and take responsibility for the paper.

A Coefficientwise verification of the polynomial pullback

This appendix verifies that the compact design (2.5)–(2.6) pulls back to the polynomial map (2.14). The calculation treats one generic coefficient g_k at a time; it is not inferred from low-degree examples.

From (2.3) and (2.4),

$$\beta(\Pi, S) = 2\frac{g_2}{g_1}\Pi + \left(3\frac{g_3}{g_1} - 1\right)\Pi S + \sum_{k=4}^N k\frac{g_k}{g_1}\Pi^k S^{k-2}. \quad (\text{A.1})$$

On the source chart,

$$S = \frac{x}{t}, \quad \Pi = tq, \quad Q = y + xq,$$

and therefore, for every $k \geq 4$,

$$\Pi^k S^{k-2} = t^2 x^{k-2} q^k. \quad (\text{A.2})$$

Substitution now gives

$$\begin{aligned} Q + \beta(\Pi, S) &= y + xq + 2\frac{g_2}{g_1}tq + \left(3\frac{g_3}{g_1} - 1\right)xq + \sum_{k=4}^N k \frac{g_k}{g_1} t^2 x^{k-2} q^k \\ &= y + 3\frac{g_3}{g_1}xq + 2\frac{g_2}{g_1}tq + \sum_{k=4}^N k \frac{g_k}{g_1} t^2 x^{k-2} q^k, \end{aligned}$$

which is the displayed coordinate B .

For the third coordinate, write

$$C_* = \frac{2G_\Pi(S)}{g_1} - BS^2.$$

Using $B = Q + \beta(\Pi, S)$ and the definition of β , we obtain

$$C_* = 2S - QS^2 + \left(1 - \frac{g_3}{g_1}\right)\Pi S^3 - \sum_{k=4}^N (k-2) \frac{g_k}{g_1} \Pi^k S^k. \quad (\text{A.3})$$

The terms with $k \geq 4$ already have the desired source form, since

$$\Pi^k S^k = (xq)^k. \quad (\text{A.4})$$

It remains to simplify the cubic part. Put $r = g_3/g_1$. Then

$$\begin{aligned} 2S - QS^2 + (1-r)\Pi S^3 &= \frac{2xt - x^2y - rx^3q}{t^2} \\ &= \frac{x(t+1) - rx^3q}{t^2}, \end{aligned}$$

where the second line uses $xy = t - 1$. Since

$$q = t^2z + \frac{1}{r}y^2(1+3t),$$

this becomes

$$-rx^3z + \frac{x(t+1 - (t-1)^2(1+3t))}{t^2}.$$

The elementary identity

$$t+1 - (t-1)^2(1+3t) = t^2(5-3t) \quad (\text{A.5})$$

therefore gives

$$2S - QS^2 + (1-r)\Pi S^3 = x(5-3t) - \frac{g_3}{g_1}x^3z.$$

Together with (A.3) and (A.4), this is exactly the displayed coordinate C .

We derived the identities after inverting t , but both sides are polynomials in $K[x, y, z]$. Localization at the nonzero polynomial t is injective, so the identities hold everywhere. At fixed Π , their differential has determinant

$$\det \frac{\partial(B, C)}{\partial(S, Q)} = -2 \left(\frac{G'_\Pi(S)}{g_1} - BS \right) = -2D.$$

The source-chart determinant is D^{-1} , and the chain rule gives $\det DF_G = -2$. This is the complete cancellation: each coefficient g_k enters independently, so the same calculation works in every degree.

B Verification and Lean correspondence

The table records the public endpoint declarations corresponding to the formalized characteristic-zero specializations used in the paper. The supplied-presentation theorem itself is stated over every field of characteristic different from 2, but the current end-to-end Lean files retain the stronger typeclass assumption `CharZero`. Several rows list more than one declaration because the paper packages construction, naturality, rank, and geometric degree into a single result. Auxiliary implementation lemmas are omitted.

Paper statement	Lean declaration
General localized quadratic-gauge fiber theorem, Theorem 3.2, and its squarefree corollary, Corollary 3.3	<code>generalGaugeLocalizedJacobianOneRepresentingEquiv</code> <code>generalGaugeLocalizedJacobianOneRepresentingEquiv_natural</code>
Jacobian-one effective form, Corollary 3.4	<code>jacobianDet_generalGaugeJacobianOneMap</code> <code>generalGaugeJacobianOneMap_totalDegree</code>
Polynomial realization, Theorem 4.1, and its effective and functorial form, Corollary 4.2	<code>automaticRealization_pageOne</code> <code>automaticRealizationFunctionField_finrank</code> <code>automaticRealizationFunctionFieldComparison</code> <code>automaticJacobianOneFiberRepresentingEquiv_natural</code>
Abstract finite-étale realization, Corollary 4.4	<code>finiteEtalePresentation</code> <code>abstractFiniteEtaleFiberRepresentingEquiv_natural</code> <code>abstractFiniteEtale_pageOne</code>
Compatibility with extension of scalars, Proposition 4.6	<code>realizationMapTarget_map</code> <code>adjoinRootBaseChangeEquiv</code>
Explicit arithmetic fiber, Section 5	<code>ExplicitQuintic.integralFiberRepresentingEquiv_natural</code> <code>ExplicitQuintic.p5_quotient_rank</code> <code>ExplicitQuintic.p5_quotient_etale</code> <code>ExplicitQuintic.integralFiberPoint_hasse_certificate</code>

The page-one certificate exposes geometric degree both through the convenience definition `automaticRealizationGeometricDegree` and, independently of that definition, through the direct standard-object equality

$$\text{finrank}_{K(\Pi, B)(C)} K(x, y, z) = \deg P.$$

Its Lean type uses `Module.finrank`, `RatFunc`, and the fraction field of the three-variable polynomial ring directly. The algebra structure is the pullback by the three displayed target coordinates, and `automaticRealizationFunctionFieldComparison` is the explicit algebra equivalence with the corresponding inverse-root extension. The certificate also includes the target denormalization, so the fixed Jacobian-one output normalization does not alter the generated target field.

The formal development already identifies the cubic coefficient of $P(a + S) - P(a)$ with the third Hasse derivative. Removing the blanket `CharZero` assumptions from the general gauge, reconstruction, function-field, and supplied-base-change files remains a formalization task; no additional ordinary mathematical argument is used in the paper.

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